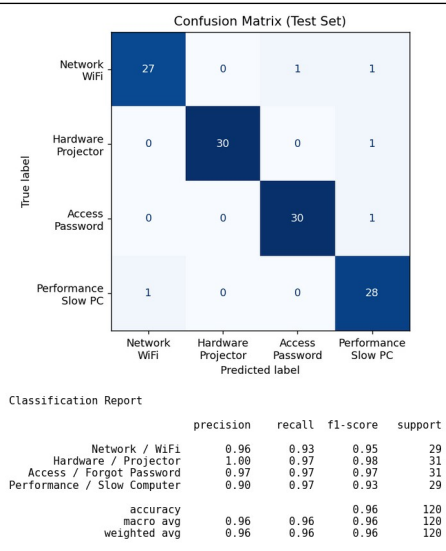
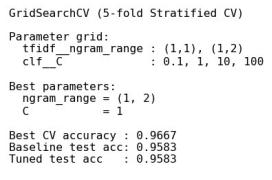


Project Development Phase Model Performance Test

Date	04-10-2026
Team ID	SWTID-2026-9856
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Classification Model (Multinomial text classification - TF-IDF + Logistic Regression): Dataset: 600 tickets (Train 480 / Test 120); 4 classes - Network/WiFi, Hardware/Projector, Access/Forgot Password, Performance/Slow Computer. Confusion Matrix - rows = actual, columns = predicted (order: WiFi, Projector, Password, Slow PC) <pre>[27 0 1 1] [0 30 0 1] [0 0 30 1] [1 0 0 28]</pre> Accuracy Score - 95.83% (0.9583) Classification Report - Macro avg Precision 0.96, Recall 0.96, F1-score 0.96; per-class F1: WiFi 0.95, Projector 0.98, Password 0.97, Slow PC 0.93. Regression Model: Not applicable (problem is classification).	 The screenshot displays the Confusion Matrix and Classification Report for the model. The Confusion Matrix shows the relationship between True Labels (Network WiFi, Hardware Projector, Access Password, Performance Slow PC) and Predicted Labels. The Classification Report provides a summary of performance metrics including precision, recall, f1-score, and support for each class, as well as overall accuracy and weighted averages.
2.	Tune the Model	Hyperparameter Tuning - GridSearchCV over TF-IDF ngram_range {(1,1),(1,2)} and Logistic Regression C {0.1, 1, 10, 100}. Best: ngram_range=(1,2), C=1. Validation Method - 80/20 stratified hold-out split plus 5-fold Stratified Cross-Validation on the training set (best CV accuracy 96.67%). Baseline test accuracy 95.83% vs tuned 95.83%.	 The screenshot shows the results of GridSearchCV for hyperparameter tuning. It lists the parameter grid used, the best parameters found (ngram_range=(1,2) and C=1), and the corresponding best CV accuracy (0.9667), baseline test accuracy (0.9583), and tuned test accuracy (0.9583).